

GNSS

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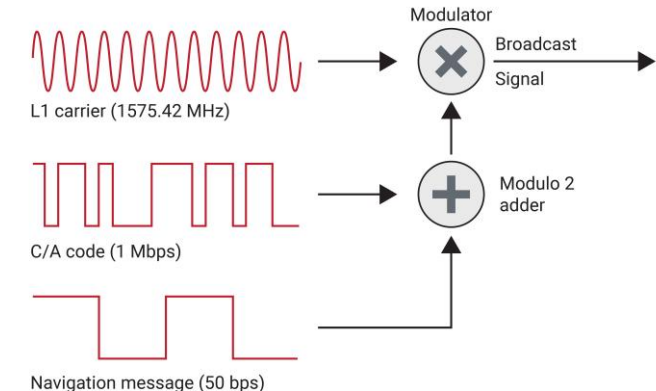
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Material adapted from

1. Stanford EE259 (Reza Nasiri Mahalati)
2. ESA's GNSS data processing book

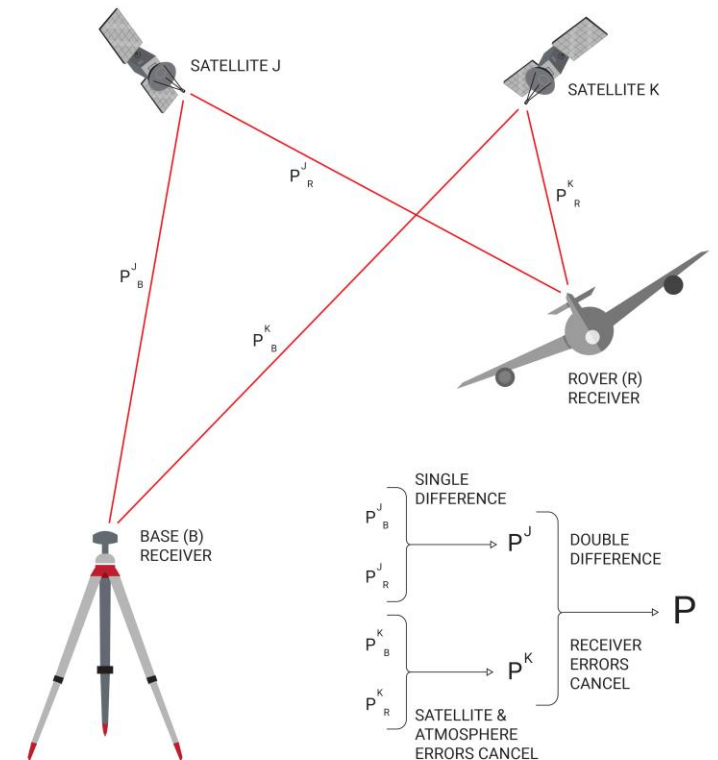
Signals

- GPS carrier frequencies: L1 (1575 MHz), L2 (1227 MHz), L5 (1176 MHz)
 - Ionospheric delay is smaller for L-band of the spectrum (1-2 GHz); no need for beam antenna for frequencies < 2 GHz
- Signal components:
 - pseudo random noise (PRN) code: unique **identifier** for each sat
 - Coarse/acquisition code (C/A): 1.023 Mbps on L1
 - Precise code (P-code): 10.23 Mbps on L1 and L2
 - **nav message** (binary): GPS time, sat health, ephemeris, and almanac data.
 - Carrier: PRN + Nav message modulated high frequency signal
- Nav message is slow: 50 bps
 - Message in 25 frames (12.5 mins for transmission)
 - Every frame
 - GPS time, sat health, ephemeris, and almanac data
 - Allows for position fix before entire nav message



Differential GPS | RTK

- Single GPS receiver: bad positioning accuracy
 - One well known GPS location: **base station**
 - Not well-known receiver but close to the base station: **rover**
 - Can correct for clock errors, orbit errors, atmospheric propagation errors | Pseudo range correction
 - DGPS: 1 meter level accuracy
- Real-time Kinematic (RTK)
 - multiple-receiver system: base receiver and rover receiver
 - carrier phase observables rather than the pseudo-range
 - Centimeter-level positioning accuracy
- Double-differencing
 - Carrier phase measurements: tracking partial wave between sat and receiver
 - Need integer number of wavelengths for distance: difficult to determine -> integer ambiguity
 - Double-differencing between two receivers: <2-cm positioning accuracy
 - Needs two sats and two receivers



RTK Double-Differencing

Intuition

- Fundamental observable: **Time** ($T_x \rightarrow R_x$)
 - Apparent distance = $c \times t \rightarrow$ Pseudorange
 - 2D analogy: ship location ([page 2](#))
- Clock synchronization error?
 - For an offset dt , range error: $v \times dt$
 - Pseudorange (with unknown dr): $R = r + dr$
 - In the lighthouse example, if R_1 and R_2 are measured with the same clock, the possible ship positions trace branch of a hyperbola
 - $R_1 - R_2 = r_1 - r_2 = \text{constant}$ (for any dt)

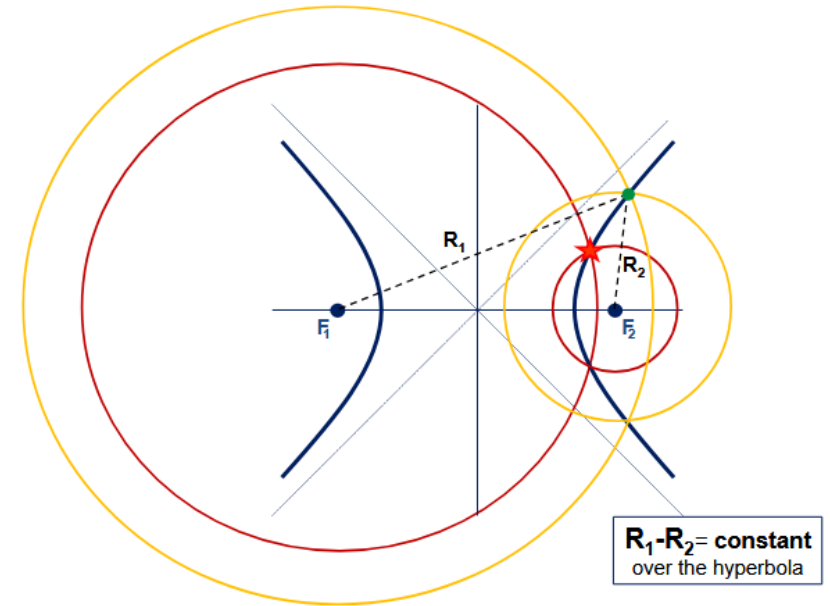
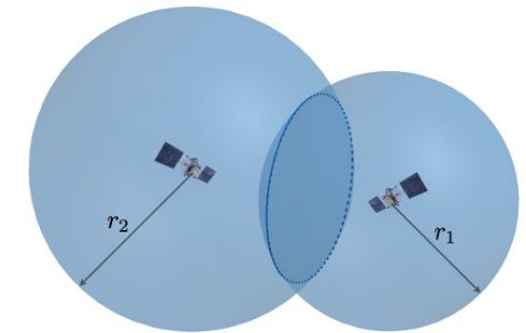
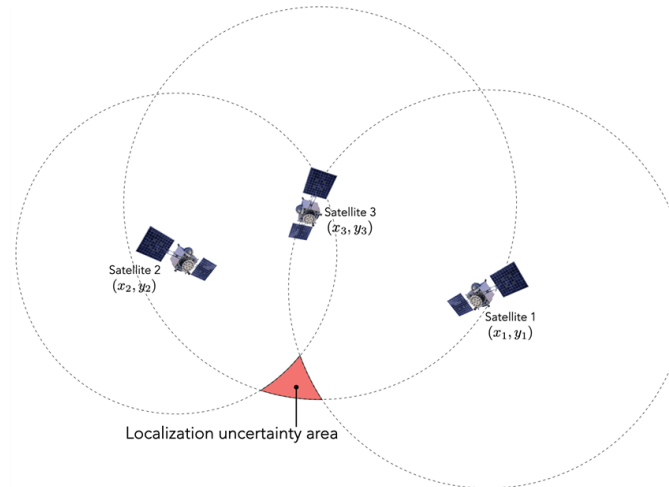


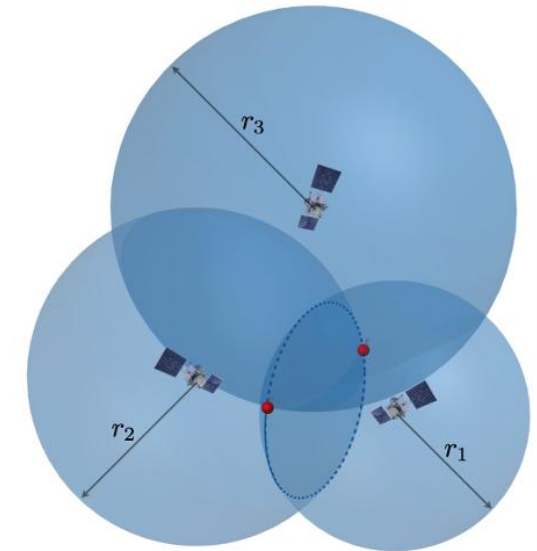
Figure 1.2: An (unknown) offset $d\tau$ in the ship's clock produces a shift in the measured ranges ($R_i = \rho_i + dr$) (or pseudoranges), varying the radius of the circle. But as both pseudoranges R_1 and R_2 have been measured with the same clock, this offset cancels the difference of ranges $R_1 - R_2 = \rho_1 - \rho_2 = \text{constant}$. Therefore, the ship is located at a branch of the hyperbola $R_1 - R_2 = \text{constant}$.

Intuition

- Position and range of a satellite limits the RX location to a sphere
 - Spheres from three satellites intersect at two possible RX locations
 - Can disambiguate by picking the location closest to earth surface
-
- GPS clocks are accurate, receivers have offset.
 - TOF measurements have clock offset $\rightarrow$ range errors
 - Receivers: jointly estimate 3D location and offset relative to GPS clock
 - Needs four sats to estimate (x, y, z, t_d)
 - Receiver: location of sat, range from sat
 - Satellite locations: ephemeris data
 - Range: ranging codes



Two satellites localize a circle



Three satellites localize two points

Signals

- Ranging signals: range from a sat to user
 - Psuedorange noise (PRN) binary codes
 - Carrier modulation using BPSK
 - IQ quadrature
 - Satellites using the same frequency: distinguished by using different ranging codes: code division multiple access (CDMA)
- Navigation signals: position of each satellite

- ▶ Signal at the receiver can be modeled as:

$$s(t) = \sqrt{2P_I}x_I(t)D_I(t)\cos(2\pi f_c t + \phi) + \sqrt{2P_Q}x_Q(t)D_Q(t)\sin(2\pi f_c t + \phi)$$

P_I : in-phase signal power (W)

P_Q : quadrature signal power (W)

$x_I(t)$: in-phase ranging code

$x_Q(t)$: quadrature ranging code

$D_I(t)$: in-phase navigation signal

$D_Q(t)$: quadrature navigation signal

f_c : carrier frequency (Hz)

ϕ : carrier phase offset (rad)

[GNSS Signals - GNSS-SDR](#)

Question: How multiple sats can transmit at the same frequency?

Answer: Autocorrelation of PRN codes at the receiver. PRNs are almost orthogonal.

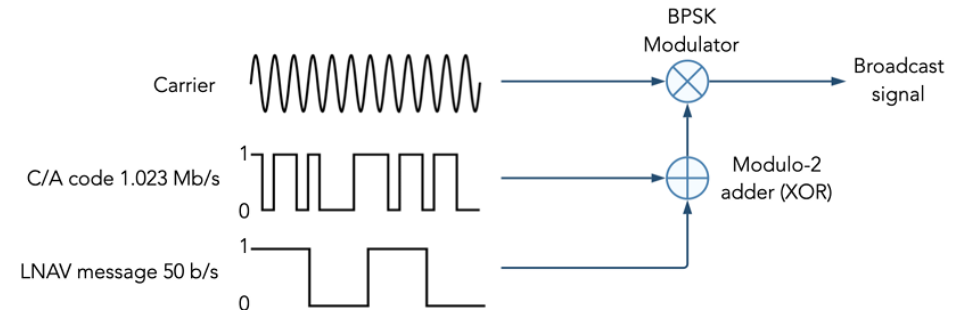
DSP at the receiver

- At the receiver antenna $r(t) = s_1 + s_2 + s_3$
- Magic in the DSP: correlation between local stored PRN code to uniquely identify each satellite
- $r(t) \cdot x_1(t) = [s_1(t) + s_2(t) + s_3(t) + n(t)] \cdot x_1(t)$
- Locked: delay locked loop (DLL) to track the incoming signal code sequence (chips)

Ranging and Nav signals

Ranging codes

- C/A coarse/acquisition code
 - Short at 1 Mbps, repeats every 1 millisc
- P(y) precision code
 - Long code at 10 Mbps, repeats every 37 weeks



Navigation signals (legacy LNAV)

- GPS date, time and sat status
- Ephemeris: satellite location
- Almanac: status and orbit information of every satellite in the constellation

- LNAV: time, status, ephemeris, almanac
- modulo-2 adder (XOR) for mixing LNAV and PRN
- BPSK for mixing with carrier

- LNAV is modulated onto C/A and P(Y) codes at 50 b/s
- BPSK (phase shift keying)
- modulo 2 ($A+B \bmod 2 == \text{XOR}$)
- 18-36 seconds for ephemeris (valid for up to 4 hours)
- Almanac (12.5 min) valid for a week

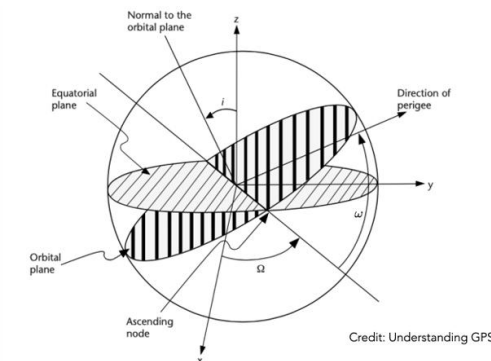
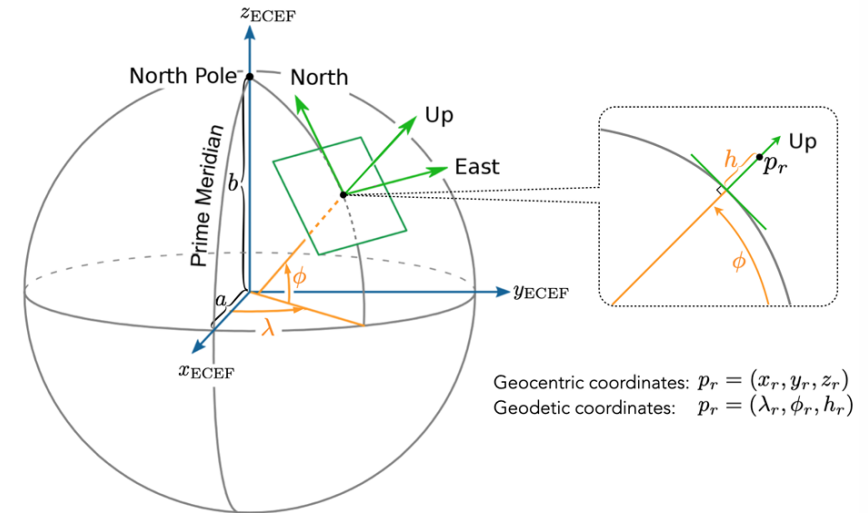
- Codes are orthogonal for better SNR in cross correlation
- Ex: code A $[1 \ 1 \ -1 \ -1]$ x code B $[1 \ -1 \ 1 \ -1]$
- Cross correlation sum: $1-1-1+1 = 0$ | Auto-corr sum: 4

- Many other options: L5, CNAV-2, L1C (I/Q)
- Better data rates, faster corrections, higher power
- Increasing data rate -> signal bandwidth increases (spectral analysis).

Coordinate Systems

- Receiver's location in ECEF frame (x,y,z) (to estimate)
- Converted to Geodetic polar frame (lat, long, height)
- Conversion using Earth model (WGS84)
- Receiver calculates sat location
- Orbit calculations in ECI frame -> converted to ECEF
- Two-body Keplerian perturbed equations of motion
- Receiver estimates sat position from orbital parameters between nav messages -> integration is too complex
- Six orbit parameters: a,e,l,w,Omega, t
 - t (epoch time or time of perigee)
 - 3-1-3 Euler angles to describe orientation of the orbit w/ inertial
- Correction factors calculation from t0

Satellite locations in orbit are calculated from the ephemeris data
Satellite ranges (TOF) are calculated from the ranging codes



ECEF: rotating cartesian frame, aligned to equatorial plane,
{x: prime meridian, y: 90 E longitude, z: geo north pole}

Ranging using PRN

ToF estimation

- The best estimator is one that maximizes the cross-correlation between received and transmitted signals.
- Correlation peak index is the best MMSE est of ToF.
 - This is the matched filter detector
- Good autocorrelation properties are required
 - PRN sequences

PRN sequences – very small cross-correlation

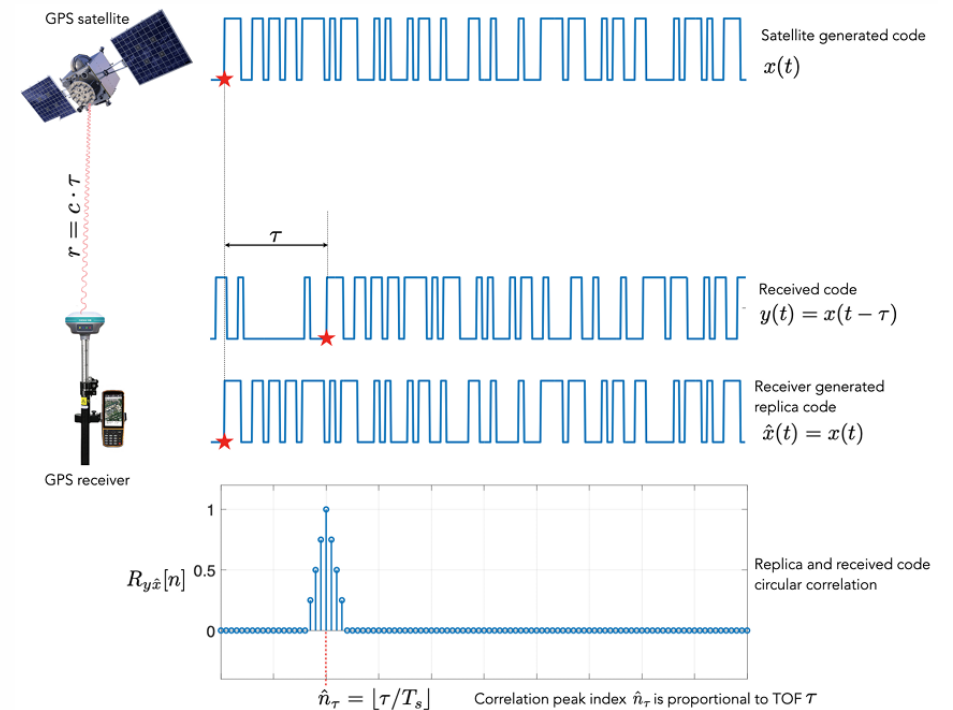
- Maximum length sequence (MLS) generated by feedback shift registers
- C/A code is an MLS of length $N = 1023$

PRN ranging

- GPS transmits $x(t)$ on carrier f (receiver knows these)
- Demodulates to recover $x(t-T)$ -> has a delay T
- Clock synchronized ($t_d = 0$)? Estimate ToF from cyclic correlation of replica code x and received code y .
 $\tau = n \cdot T_s \rightarrow$ calculate range $r = c \cdot \tau$

- ▶ Given $y[k]$ and $x[k]$, want to estimate attenuation α , and TOF n_τ
- ▶ The least-squares estimator of α and n_τ minimizes the MSE error between the received signal and a scaled and delayed version of $x[n]$:

$$\hat{\alpha}, \hat{n}_\tau = \underset{\alpha, n}{\operatorname{argmin}} \sum_k (y[k] - \alpha x[k - n])^2$$



Very small cross-correlation between codes from different sats enables all satellites to use the same carrier frequencies (L1- L5) without interference. Nearly-orthogonal.

Ranging using PRN

ToF estimation

- In practice, clock offset exists
- Doppler shift in carrier frequency due to relative motion between sat and Rx ($f_d = \frac{\Delta \mathbf{v} \cdot \hat{\mathbf{r}}}{c} f_c$)
- Rx local oscillator needs to “lock in” to the shifted carrier frequency for demodulation

✓ Satellite ranges (TOF) are calculated from the ranging codes

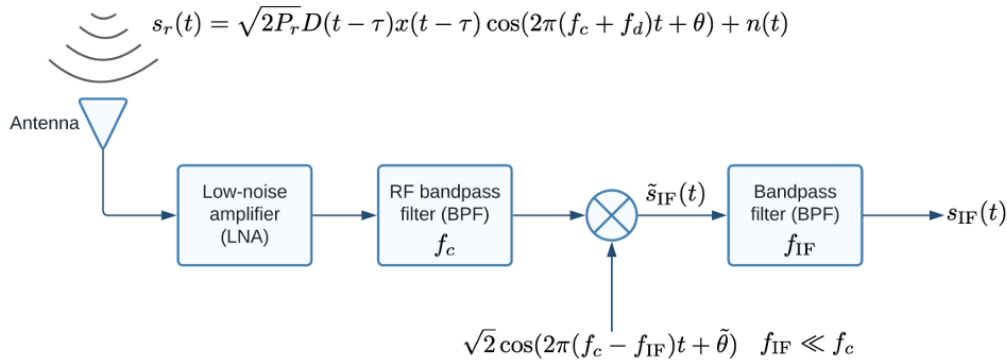
Range ambiguity

- Cyclic autocorrelation of the PRN code: periodic correlation peaks, which one is range?
- Pick correlation index by eyeballing!
 - C/A code: Δr for a ΔT could be approx 300 km
- requires sophisticated algos.

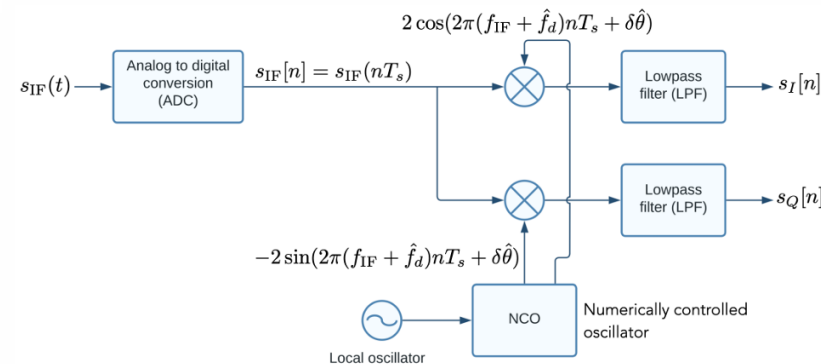
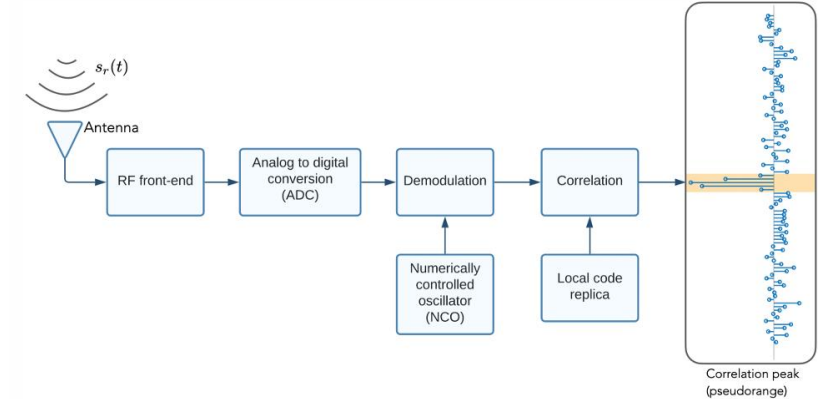
Receiver architecture

Demodulation

- Finds, tracks, and cancels Doppler shift ‘
 - Closed loop feedback control of numerically controlled oscillator
- Correlation, finds and tracks ToF ($\tau + \tau_d$)
- Front-end – heterodyne receiver
 - Brings carrier frequency to intermediate frequency



Nav message bitrate is small -> ToF is tracked from correlation of PRN codes.



Downconvert carrier to IF (analog) -> correct for Doppler – quadrature demod (digital)
All the signal power is in in-phase component.

Why is IF important: all the filters, amplifiers can be designed for IF frequency. Standardized components.

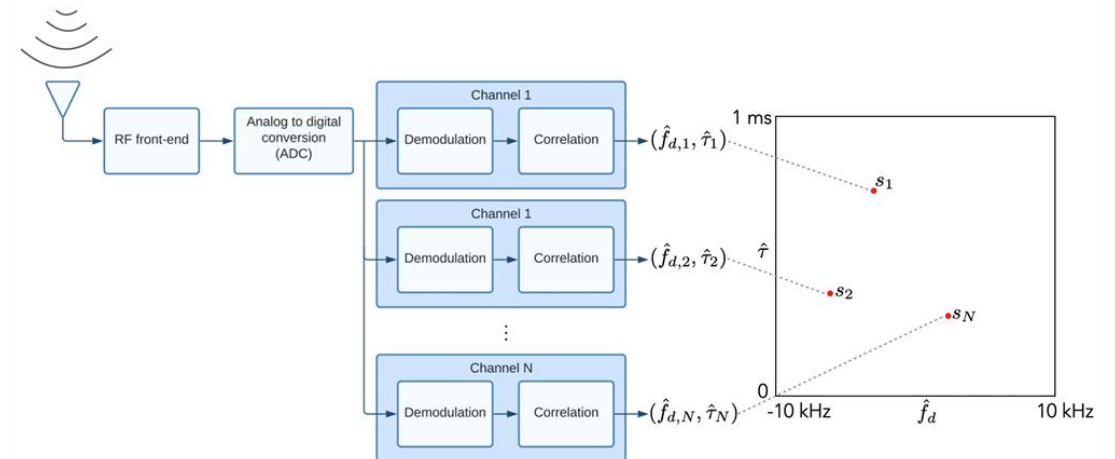
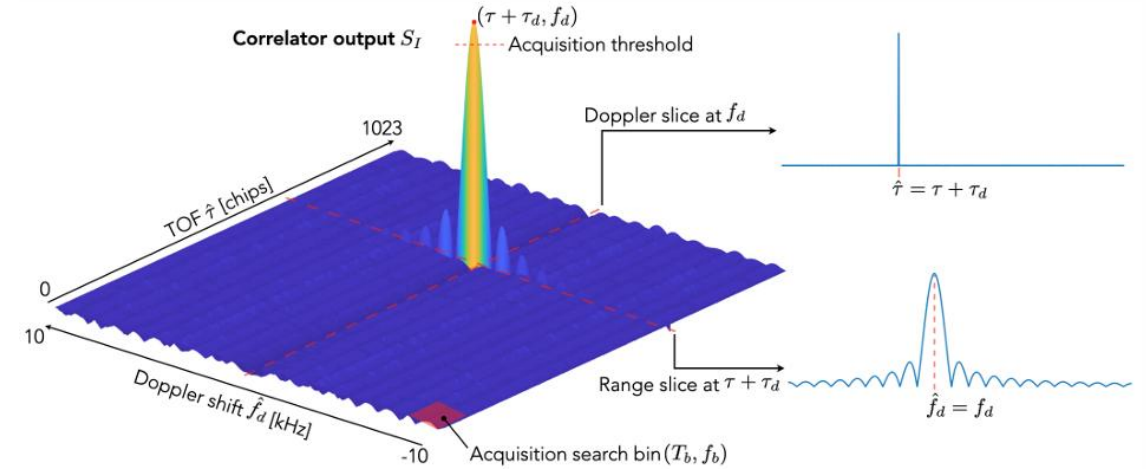
Acquisition and tracking

Acquisition

- Rx estimates $(f_d, \tau + \tau_d)$ for multiple sats in view
- 2D search over Doppler frequencies and ToF
- 5-10 seconds for C/A codes

Tracking

- Rx has multiple channels for simultaneous acquisition and tracking
- Each channel's correlator has ranging code for specific sat.
- Switches to tracking after acquisition, variations in Doppler and ToF are tracked.
- Varying Doppler and ToF via feedback control from correlator to NCO.



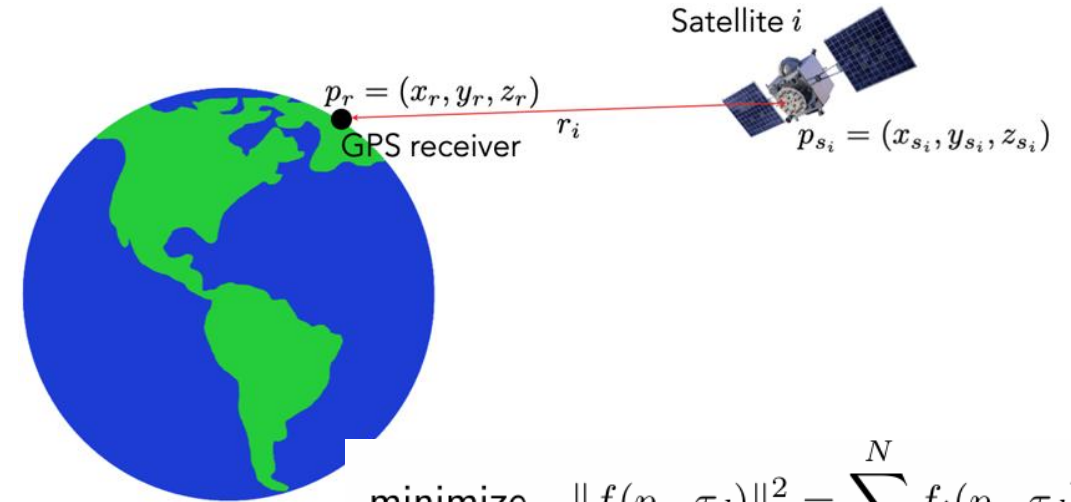
Position and Velocity Estimation

Pseudorange measurements

- I have pseudorange measurements, estimate Rx pos, clock offset

$$\rho_i = r_i + c\tau_d$$

- Four unknowns, require 4 simultaneous pseudorange measurements to recover Rx position and ToF.
- Nonlinear equation: Gaussian Nonlinear Least Squares with Differential Correction
 - Needs Jacobian, initial guess, iterate until convergence.
 - Using previous position as initial guess.



$$\text{minimize } \|f(p_r, \tau_d)\|^2 = \sum_{i=1}^N f_i(p_r, \tau_d)^2$$

$$\begin{aligned} f_i(p_r, \tau_d) &= \rho_i - r_i - c\tau_d \\ &= \rho_i - \|p_r - p_{s_i}\| - c\tau_d \\ &= \rho_i - \sqrt{(x_r - x_{s_i})^2 + (y_r - y_{s_i})^2 + (z_r - z_{s_i})^2} - c\tau_d \end{aligned}$$

Velocity Estimation

- Satellite velocity from ephemeris
- Rx velocity from Doppler shifts

$$f_{d,i} = \frac{\Delta v_i \cdot \hat{r}_i}{c} f_c = \frac{(v_r - v_{s_i}) \cdot (p_r - p_{s_i})}{c \|p_r - p_{s_i}\|} f_c$$

Complexity: Pseudorange and Doppler measurements errors from multipath interference, ionospheric delays, ephemeris errors, EMI

GNSS+INS: EKF

NED frame: velocities and relative positions. Origin: reference/current position.

N: Points to true North parallel to line of latitude

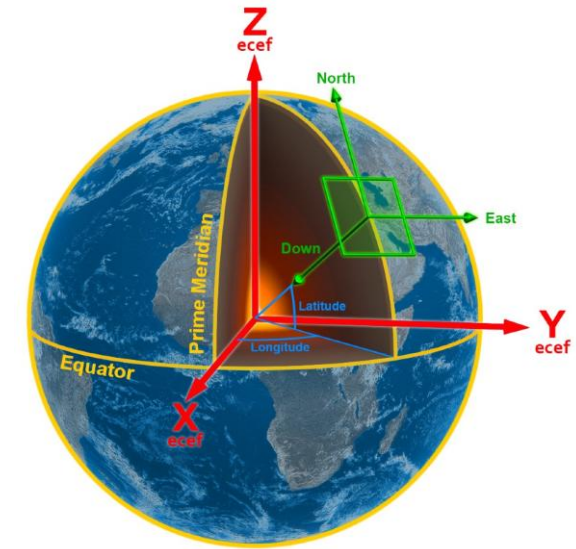
E: perpendicular to N and points to line of longitude at the point

D: center of Earth

(or) equivalently ENU frame.

- Convert GNSS geodetic coordinates to ENU using `GeographicLib::LocalCartesian`

- EKF – tightly coupled GNSS + IMU sensor fusion
- Vehicle state estimation (position, velocity, attitude)
- Propagate kinematics – IMU at 100 Hz, filter update with GNSS measurements at 10 Hz
- ROS2 C++ implementation and notes: <https://github.com/ram-bhaskara/GNSS-IMU-ErrorStateEKF>



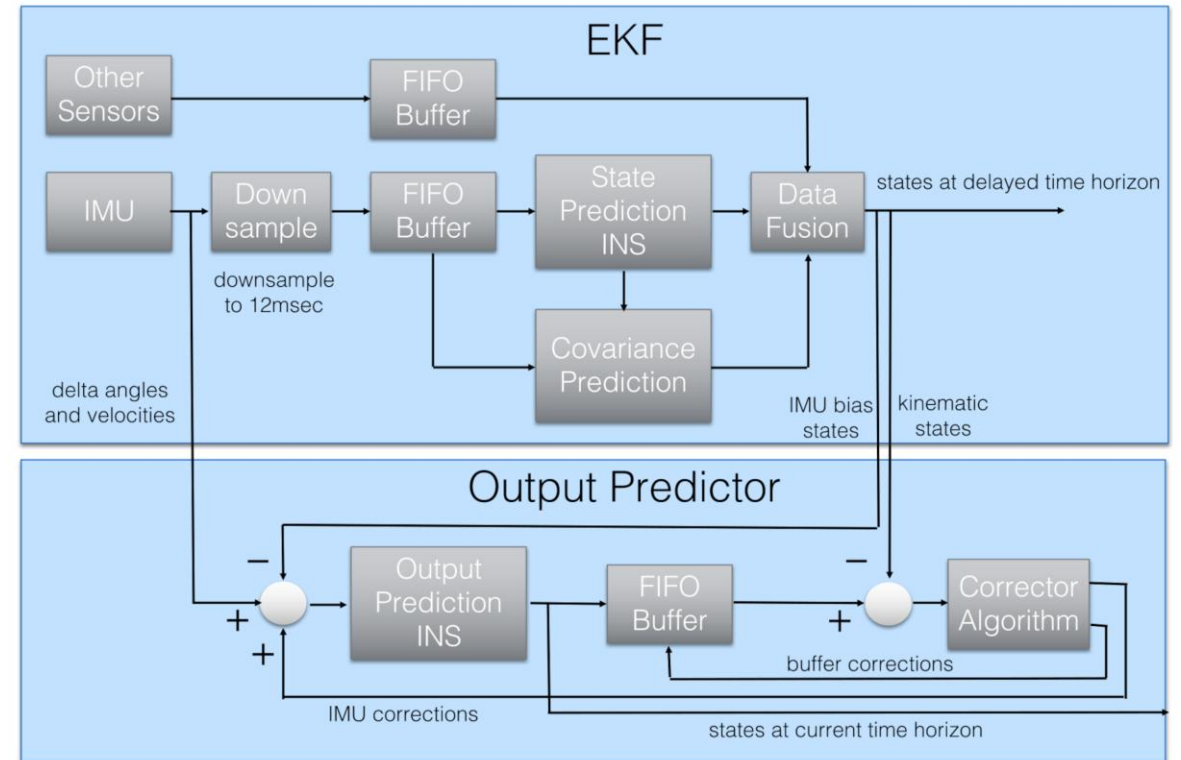
Transformation DCM from ECEF->NED:
Rotate about Z_{ecef} through longitude (λ) -> rotate about local y by latitude (ϕ)

Tightly coupled: get the GNSS data (LLA) instead of pos, vel (loosely coupled) to estimate the state.

GNSS+INS: EKF

EKF:

1. Downsample IMU data to improve SNR
2. Prediction from IMU
3. Correction from GNSS



Tightly coupled: get the GNSS data (LLA) instead of pos, vel (loosely coupled) to estimate the state.